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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2025

CS2101-1 - COMPUTER ORGANIZATION & ARCHITECTURE
(FOR CSE, CYB)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is computer organization?**
A) Structure and behavior of a computer system as observed by the user
B) Structure of a computer system as observed by the developer
C) Structure and behavior of a computer system as observed by the developer
D) All of the mentioned
2. **Which architecture is power efficient**
A) RISC
B) ISA
C) IANA
D) CISC
3. **Small, extremely fast RAMs are called _____.**
A) Heaps
B) Accumulators
C) Stacks
D) Cache
4. **During the execution of a program, which gets initialized first?**
A) MAR
B) MDR
C) PC
D) IR
5. **What does the mantissa represent in a normalized IEEE single precision floating -point number?**
A) The whole number part
B) The fractional part, excluding the hidden bit
C) The exponent
D) The sign bit
6. **Booth's algorithm is primarily used for which type of multiplication?**
A) Unsigned multiplication
B) Signed multiplication
C) Fractional multiplication
D) Complex number multiplication
7. **In restoring division, what happens if the subtraction results in a negative value?**
A) Restore the original value
B) Perform another subtraction
C) Shift left
D) Multiply the quotient
8. **In a direct mapped cache, which replacement algorithm is most suitable?**
A) Least Recently Used (LRU)
B) First In, First Out (FIFO)
C) Random Replacement
D) No replacement needed
9. **In a single bus architecture, when multiple devices request access to the bus simultaneously, which mechanism is primarily used to control access and prevent conflicts, and what is the main impact of this mechanism on system performance?**
A) Polling; it increases the data transfer rate by managing access efficiently.
B) Interrupts; it ensures prioritized access, improving overall response time.
C) Bus Arbitration; it serializes access requests, which can cause delays and reduce overall throughput.
D) Direct Memory Access (DMA); it allows all devices to access memory directly without conflicts.



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10. The circuit used to store one bit of data is known as
A) Register B) Encoder
C) Flip Flop D) Decoder
11. Which register is connected to the MUX?
A) Y B) Z
C) R0 D) Temp
12. The main virtue for using single Bus structure is _____
A) Fast data transfers B) Cost effective connectivity and speed
C) Cost effective connectivity and ease of attaching peripheral devices D) None of the mentioned
13. In a three BUS architecture, how many input and output ports are there?
A) 2 output and 2 input B) 1 output and 2 input
C) 2 output and 1 input D) 1 output and 1 input
14. In which protocol are both the cache and the main memory updated during a Write operation?
A) Write-back protocol B) Load-through protocol
C) Write-through protocol D) Early restart protocol
15. When dealing with multiple devices interrupts, which mechanism is easy to implement?
A) Polling method B) Vectored interrupts
C) Interrupt nesting D) None of the mentioned
16. The code sent by the device in vectored interrupt is _____ long.
A) upto 16 bits B) upto 32 bits
C) upto 24 bits D) 4-8 bits
17. An interrupt breaks the execution of instructions and diverts its execution to
A) Interrupt service routine B) Counter word register
C) Execution unit D) control unit
18. Interrupts initiated by an instruction is called as _____
A) Internal B) External
C) Hardware D) Software
19. SCSI stands for _____
A) Signal Computer System Interface B) Small Computer System Interface
C) Small Coding System Interface D) Signal Coding System Interface
20. The transfer rate, when the USB is operating in low-speed of operation is _____
A) 5 Mb/s B) 12 Mb/s
C) 2.5 Mb/s D) 1.5 Mb/s

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the major functional units of a computer system with a neat block diagram.	6	L*2	1	1
	b) Explain three-address and two-address assembly code for the expression $Z = (M - N) + (P * Q)$,	6	L2	1	1
	c) Construct the sequential multiplication algorithm steps, multiply 11 by 13 (with 13 as the multiplier).	4	L3	1	1
2.	a) Explain the connection between the processor and memory with a suitable diagram. Outline the various registers used in the processor during instruction execution,	6	L2	1	1
	b) With a neat diagram, explain straight-line sequencing of instructions.	6	L2	1	1
	c) Using the Restoring division method divide 14 by 3.	4	L3	2	1

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|------|---|---|----|---|-----|
| 3 a) | Explain the following addressing modes with syntax and example
i, Indirect Addressing mode
ii, direct Addressing mode
iii, Immediate Addressing Mode | 6 | L2 | 2 | 1,2 |
| b) | Explain the circuit arrangement and outline the steps involved in sequential multiplication of unsigned numbers with a neat diagram. | 6 | L2 | 2 | 1 |
| c) | Construct the floating number (1259.125) in a single and double precision format. | 4 | L3 | 2 | 1 |

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | Explain the Single bus organization of the data path inside a processor. | 6 | L2 | 3 | 1 |
| b) | Explain asynchronous DRAM and the internal organization of 2Mx8 dynamic memory chip. | 6 | L2 | 3 | 1,2 |
| c) | Explain the different types of ROM. | 4 | L2 | 3 | 1 |
| 5. a) | Explain about Three-bus organization of the data path. Provide control sequence for ADD R4, R5, R6. | 6 | L2 | 3 | 1 |
| b) | Illustrate and explain the internal organization of a 128-bit RAM structured as a 16x8 cell array, detailing its key components and operation with a diagram. | 6 | L2 | 3 | 1 |
| c) | Illustrate the differences between hardwired and microprogrammed control. | 4 | L2 | 4 | 1 |
| 6. a) | Explain the concept of Hard wired control with a neat diagram. | 6 | L2 | 4 | 1 |
| b) | Explain the direct mapping and associative mapping with the diagram. | 6 | L2 | 4 | 1 |
| c) | Explain address translation with a neat diagram. | 4 | L2 | 4 | 1 |

Unit – III

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|-------|---|---|----|---|-----|
| 7. a) | Define interrupts and explain the interrupts nesting and vectored interrupts with a neat diagram. | 8 | L2 | 5 | 1 |
| b) | Explain centralized and distributed bus arbitration with a neat diagram. | 8 | L2 | 5 | 11 |
| 8. a) | Explain the direct memory access with a neat diagram. | 8 | L2 | 5 | 1,2 |
| b) | Explain the USB and PCI bus with a neat diagram. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
